



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
Chennai- 602105



Student Name: Dinnepati Sindhu Prasad

Reg. No.: 192311271

Course Code: CSA-1522

Slot: C

Course Name: Cloud Computing & Big Data Analytics

Course Faculty: Dr.P.Rajaram

Project Title: Hadoop Streaming-Based Real-Time Weather Data Analytics System

Module Photographs:

Module 2 – Fuel Consumption Analysis and Energy Efficiency Optimization

Inputs:

- Temperature
- Humidity
- Rainfall
- Wind Speed
- Pressure

Output:

- Average Temperature
- Rainfall Statistics
- Humidity Analysis
- Weather Trend Report

Techniques Used:

- Hadoop Streaming
- MapReduce
- Statistical Analysis

Process:

- Read weather data from HDFS.
- Execute Mapper using Hadoop Streaming.
- Aggregate results using Reducer.
- Analyze weather trends.

Mevalurkuppam, Tamil Nadu, India
Ece Department, Mevalurkuppam, Tamil Nadu
602105, India
Lat 13.026046° Long 80.01613°
Tuesday, 04/09/2020 10:57 AM GMT +05:30

```
main.py
22 for row in data:
23     mapped["Temperature"].append(row[1])
24     mapped["Humidity"].append(row[2])
25     mapped["Rainfall"].append(row[3])
26     mapped["Wind Speed"].append(row[4])
27     mapped["Pressure"].append(row[5])
28
29 print("MAPPER OUTPUT")
30 print("-----")
31
32 for key in mapped:
33     print(key, ":", mapped[key])
34     print("\nREDUCER OUTPUT")
35     print("-----")
36
37 for key in mapped:
38     values = mapped[key]
39
40     average = sum(values) / len(values)
41     minimum = min(values)
```

Project Description:

The project “Hadoop Streaming-Based Real-Time Weather Data Analytics System” aims to provide an efficient approach for processing and analyzing large-scale weather data for effective monitoring and decision-making. Module 2 focuses on Weather Data Analysis Using Hadoop Streaming, which is the core processing stage responsible for converting stored weather data into meaningful statistical information and identifying important weather trends. In this module, weather parameters such as temperature, humidity, rainfall, wind speed, and atmospheric pressure are retrieved from HDFS and processed using the Hadoop Streaming framework. The Mapper processes individual weather records and extracts the required information, while the Reducer groups and aggregates the processed data to generate useful results. Statistical analysis is performed to determine values such as average temperature, rainfall statistics, and humidity levels, while trend analysis helps identify variations and patterns in weather conditions. The final outputs of this module include average temperature, rainfall statistics, humidity analysis, and weather trend reports, which are subsequently used by the monitoring system for displaying analyzed weather information and supporting timely decision-making.

Student Signature

Guide Signature